Cell Design Specification

VBF-T4R2-DS-01 | case t4_r2

Case t4_r2: extreme-cold environment equipment battery (−20 °C, 1C discharge retention ≥ 95 %, ED ≥ 327.18 Wh/kg, volumetric ED ≥ 880 Wh/L)

Cell Design Specification — VBF-T4R2-DS-01

Case t4_r2: extreme-cold environment equipment battery (−20 °C, 1C discharge retention ≥ 95 %, ED ≥ 327.18 Wh/kg, volumetric ED ≥ 880 Wh/L)

Generation date: 2026-08-26 (headless session; prepared/reviewed/approved signature block in delivery_index)

> All values below are mechanically taken from the Chen2020 parameter set (overridden by cell/params_E3.json), from tool output JSONs under cell/, or from literature defaults explicitly annotated. No number is written from memory; missing items are stated as "Not provided".

1. Basic specification

2. Electrode and separator

N/P basis (mechanical): full-window electrode capacities = c_max × (full stoichiometry window at equilibrium voltage floors) × F/3600 × thickness × ε_active × area. Values from the solved DFN (C/10 floors): anode window 0→0.9014 → 5.253 Ah; cathode window 0.2700→0.8531 → 5.092 Ah; ε_active from parameter set (neg 0.75, pos 0.665). Inventory closure check: both electrodes transfer 5.0384 Ah over the 1C discharge span (computed 5.0385/5.0387) — consistent at 0.1 %. Cell is positive-limited (cathode exhausts first at 2.5 V floor; anode retains x = 0.0276).

3. Process design parameters

4. Mass breakdown (contract-caliber, electrolyte excluded in cell mass)

Active-layer masses carry no binder/conductive-additive split (set has no such keys) — split rows appear in the BOM with literature default 96/2/2 wt% annotation.

5. Performance verification (DFN verdict grade)

Reproducibility: two independent DFN 4C runs returned identical T_max/anode-potential min (r5_E3_4c_dfn.json and _dfn2).

6. Design notes (what changed and why; cites evaluate-log reasoning)

- Problem: baseline Chen2020 passes W1/W2/W4(SPMe) but fails W3 (843.5 vs 880 Wh/L) and W5 (4C plating at DFN: anode separator-interface saturation, ap_min −0.19…−0.26 V) (evaluate R1, funnel R1).
- W3 fix (R2): stack compression — porosities 0.28/0.22, separator 12→8 µm, collectors 16/12→12/8 µm (B1 893.1 Wh/L) (evaluate R2).
- W4/W5 fix (R3–R5, escalation layer 1): geometry-only plating levers exhausted (particle radius neutral-to-worse; anode margin monotone but insufficient — funnel R3, strike 1). Lever class switched to electrolyte transport ($\sigma/D/t^+$) + cooling + anode porosity headroom: ap_min −0.259 → +0.0060 V, T_max 357.5 → 326.96 K (evaluate R4/R5). E3 = E2 (transport-max, h=80) + anode porosity 0.22→0.28, the final 0.078 V lever (funnel R5).
- Honesty procedures applied: cold-soak lowT (Initial temperature [K] 253.15), C-rate re-nominalization (E3 probe 5.0392 within ±3 % → nominal 5.0 kept), SPMe-at-4C plating indicator rejected (R2 diagnostic) — all safety verdicts DFN-grade, cell volume mechanically refreshed (1.939e-5 m^3).
- Remaining caveats: electrolyte transport scalars are formulation estimates (flat-in-T in this parameter set); Stage-5 true compute skipped (real_compute=false, endorse entry); wound outer dimensions/enclosure not provided; cycle life not simulated.

Field	Value	Source
Electrochemical system	NMC811-like positive / graphite negative, LiPF6-class electrolyte (Chen2020, LGM50-derived teaching parameterization)	Chen2020 parameter set (entry-0 default record)
Nominal capacity	5.0 Ah	Nominal cell capacity [A.h] = 5.0 (parameter set)
Measured 1C capacity (25 °C, DFN)	5.0392 Ah (+0.78 % vs nominal; re-nominalization threshold ±3 % → nominal kept)	cell/r5_E3_1c_dfn.json
Voltage window	2.5 – 4.2 V	Lower/Upper voltage cut-off [V] (parameter set)

Field	Value	Source
Cell dimensions (electrode strip)	height 65.0 mm × width 1580.0 mm × stack thickness 0.1888 mm	Electrode height/width [m], stack = 75.6+85.2+8+12+8 µm
Wound/outer cell dimensions, shell thickness	Not provided (no enclosure parameters in set)	honest "Not provided" per template
Electrolyte formulation	High-transport blended-single-ion class: σ 1.6 S/m, D 3.2e-10 m^2/s, t^+ 0.6, c_0 1000 mol/m^3. Formulation estimates (parameter bridge), temperature-independent scalars — marked estimate, not measured	params_E3.json overrides; B4/E2/E3 branch reasoning (evaluate R4/R5)
Cation transference number	0.6 (estimate)	parameter override
Cooling design	total heat transfer coefficient h = 80 W/(m²·K) (liquid-cold-plate class)	params_E3.json

Layer	Thickness	Porosity	Particle radius	Material / density	Source
Positive electrode	75.6 µm	0.28	3.5 µm	NMC811-class, 3262 kg/m³ (from layer mass ÷ (t×(1−ε)))	parameter set + calc-energy layer_kg_m2
Negative electrode	85.2 µm	0.28 (raised from 0.22 — plating headroom, +43.6 % effective transport via Bruggeman ε^1.5)	3.0 µm	graphite-class, 1657 kg/m³	parameter set + params_E3.json
Separator	8 µm	0.47	—	397 kg/m³	parameter set
Positive current collector	12 µm	—	—	Al, 2700 kg/m³	parameter set
Negative current collector	8 µm	—	—	Cu, 8960 kg/m³	parameter set
N/P ratio	1.03			basis note below	

Parameter	Value	Formula	Notes
Positive areal density	177.6 g/m²	layer_kg_m2 × 1000	t × (1−ε) × density = 75.6 µm × 0.72 × 3262 kg/m³ = 177.6 g/m² ✓
Negative areal density	101.6 g/m²	layer_kg_m2 × 1000	85.2 µm × 0.72 × 1657 = 101.6 ✓
Positive compaction density	2.35 g/cm³	density × (1−ε) ÷ 1000	3262×0.72/1000
Negative compaction density	1.19 g/cm³	density × (1−ε) ÷ 1000	1657×0.72/1000
Electrolyte fill amount	6.01 g/cell	pore volume × 1.2 g/cm³ (literature electrolyte density) × fill factor 1.0	pore volume 5.010e-6 m³ (ε-weighted layers × area); density is literature default (set has none) — annotated
Formation recommendation	0.1C CC to 4.2 V, 25 °C, 2 cycles	design recommended value; production-line value requires tuning	annotated "design recommendation"

Component	g/cell	kg/kWh	Source
Positive active layer	18.23	1.004	layer_kg_m2 × area (calc-energy)
Negative active layer	10.44	0.575	layer_kg_m2 × area
Positive CC (Al 12 µm)	3.33	0.183	layer_kg_m2 × area
Negative CC (Cu 8 µm)	7.36	0.405	layer_kg_m2 × area
Separator (8 µm)	0.17	0.010	layer_kg_m2 × area
Electrolyte (pore fill)	6.01	0.331	pore volume × 1.2 g/cm³ (literature)
Total (incl. electrolyte)	45.55	2.508	
Total without electrolyte	39.54	2.177	matches calc-energy mass_kg 0.039536

Item	Value	Threshold	Verdict	Source
1C retention, −20 °C	0.9938	≥ 0.95	✓ (+4.6 ppt)	r5_E3_retention_dfn.json (5.0078/5.0392 Ah)

Item	Value	Threshold	Verdict	Source
Energy density (gravimetric)	459.45 Wh/kg	≥ 327.18	✓ (+132.3)	r5_E3_energy_dfn.json (contract caliber, electrolyte excluded)
Volumetric energy density	936.82 Wh/L	≥ 880	✓ (+56.8)	r5_E3_energy_dfn.json (Σ layer thickness × area)
4C charge temperature rise (45 °C ambient)	T_max 326.96 K	≤ 333.15	✓ (margin 6.19 K)	r5_E3_4c_dfn2.json (stage-4 exam, second independent run)
4C plating	min anode surface potential +0.0060 V	≥ 0 (plated=false)	✓ (margin 6 mV; 0/307 points negative)	r5_E3_4c_dfn2.json anode_potential_v